# llmXive Automated Review of ResearchClawBench: A Benchmark for End-to-End Autonomous Scientific Research

*llmXive automated review panel*

**Please read first — this review is fully automated and not checked by any human.** It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

## How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

**This paper's panel (9 lenses):** Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

## Claim Accuracy

*Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several specific quantitative claims regarding benchmark results and error analysis that require verification against the provided data tables.

First, in Section 4.2, the claim that "Claude Code... wins only 14 out of 40 tasks" is a derived statistic not explicitly shown in the provided text or the truncated tables. While the total score (21.5) is consistent with the table, the "win" count (defined as the highest score per task) must be explicitly calculated and verified against the full 40-task dataset. The current text does not provide the breakdown of wins per task, making this specific number difficult to validate without the full table.

Second, the examples given for domain-specific performance in Section 4.2 ("Claude-Opus-4.6 in Astronomy, GLM-5.1 in Chemistry") appear inconsistent with the provided snippet data. For `Astronomy_000`, the table shows Codex CLI (29.4) outperforming Claude-Opus-4.6 (25.6). For `Chemistry_000`, ResearchClaw (18.25) and GLM-5.1 (18.95) are close, but the claim of GLM-5.1 leading the domain needs confirmation across all Chemistry tasks, as the snippet suggests mixed results. The authors should ensure these specific examples accurately reflect the aggregate or per-task winners.

Third, in Section 4.4, the text states that analysis of 280 runs identifies "six error types." While the text later mentions the other three types (Goal Misalignment, Reliability/Reporting Failure, Execution Failure), the initial numbered list only enumerates three. This creates a minor inconsistency in the presentation of the error taxonomy. The list should be expanded to include all six types to match the stated count.

Finally, the arithmetic for the total number of runs (7 agents $\times$ 40 tasks = 280) is correct, and the specific score for OpenClaw on `Physics_002` (27.45) matches the table. However, the claim that this is the "highest score among all autonomous agents" relies on the full table data, which is partially truncated in the provided source. The authors should confirm that no other agent (e.g., Nanobot, ARIS) scored higher in the omitted rows.

**Action items.**

- [**writing**] The paper makes several specific quantitative claims regarding benchmark results and error analysis that require verification against the provided data tables. First, in Section 4.2, the claim that "Claude Code... wins only 14 out of 40 tasks" is a derived statistic not explicitly shown in the provided text or the truncated tables. While the total score (21.5) is consistent with the table, the "win" count (defined as the highest score per task) must be explicitly calculated and verified against

## Figure Critic

*Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_figure_critic.md`

### Figure 1

The figure effectively visualizes the cost-score trade-off, but the caption is misleading by claiming to show 'runtime' data which is absent from the plot. Additionally, the internal legend box is slightly ambiguous regarding the 'Efficient knee' label.

### Figure 2

Figure 2 effectively visualizes the error type distribution using six distinct donut charts, each clearly labeled with the category name and percentage. The caption provides comprehensive definitions for each error type, ensuring the data is fully interpretable without ambiguity.

### Figure 3

The figure effectively contrasts the agent's partial output with the full rubric requirements, but the 'Agent Answer' panel is cluttered with unexplained plots and lacks the unified comparisons and specific models (Gate-Counting, N=56 MB) demanded by the rubrics, making the 'recovery' claim in the caption misleading.

### Figure 4

Figure 4 provides a clear and comprehensive schematic of the ResearchClawBench framework, effectively illustrating the workflow from task definition to agent execution and evaluation. The visual hierarchy, color-coding, and labeled components align well with the caption's description of the system's architecture.

### Figure 5

The figure effectively visualizes the data construction workflow with clear section headers, but the specific text content within the panels is dense and the diagram omits the 'rubric building' and 'task packaging' steps described in the caption.

### Figure 6

Figure 6 effectively illustrates the proposed evaluation metric using two distinct scientific case studies (Atmospheric Mechanism Analysis and Polymer Design Optimization). The visual progression from ‘No Discovery’ to ‘Discovery’ is clearly supported by specific examples, scores, and detailed rubric descriptions for each stage.

**Action items.**

- [**writing**] Figure 1: The caption mentions ‘runtime’ as a variable, but the plot only displays ‘Mean cost per task (USD)’ on the x-axis; the runtime data is missing.
- [**science**] Figure 1: The caption describes the plot as showing ‘Resource-score relationships’, but the x-axis is labeled ‘Mean cost per task (USD)’ while the y-axis is ‘Mean rubric score’; the ‘runtime’ component mentioned in the caption is not visualized.
- [**writing**] Figure 1: The legend box ‘Efficient knee: Qwen3.7’ is ambiguous; it is unclear if this refers to the star marker, the specific model, or the concept of the knee itself.
- [**science**] Figure 3: The ‘Agent Answer’ panel (left) is missing the ‘Gate-Counting Model’ plot and the ‘N=56 MB Regression’ plot required by Rubrics 3 and 4, yet the caption claims the agent ‘recovers the most direct XEB trend’ without noting these critical missing components.
- [**science**] Figure 3: The ‘Agent Answer’ panel (left) lacks a unified multi-estimator comparison plot (e.g., XEB, MB, and Gate-Counting on one axis) as required by Rubric 1, instead showing fragmented, separate plots that do not allow for direct comparison.
- [**writing**] Figure 3: The ‘Agent Answer’ panel (left) contains a plot titled ‘N=40 verification: XEB fidelity vs depth’ which is not referenced or explained in the rubrics or caption, creating confusion about its purpose.
- [**science**] Figure 5: The diagram depicts a linear workflow (Steps 1-5) but omits the ‘build rubrics’ and ‘package standardized tasks’ steps explicitly mentioned in the caption, creating a disconnect between the visual process and the described methodology.
- [**writing**] Figure 5: The text inside the numbered panels (e.g., Panel 2 ‘The paper’s key claim is...’) is dense and small, reducing legibility compared to the larger section headers.

## Jargon Police

*Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific acronyms and jargon that hinder accessibility for a broad scientific audience. In Section 3.2, the term “ReAct-style loop” is introduced without definition; while standard in AI agent literature, it should be briefly explained (e.g., “a reasoning-and-action loop”) for general readers. Similarly, Section 5.4 utilizes “XEB” (Cross-Entropy Benchmark) without expansion, which is opaque to non-quantum computing specialists.

The term “dry-lab” in Section 6 is colloquial; “computational-only” or “simulation-based” would be more precise and inclusive. In the Energy_000 case study (Section 4.4) and the case study in Section 5.4, “LHS” is used without defining it as Latin Hypercube Sampling. Furthermore, the frequent use of “rubrics” to describe evaluation criteria (Sections 3.3, 4.1, 5.1) is slightly jargon-heavy; “scoring criteria” or “evaluation checklists” would be clearer. Finally, “Volkov final-state” in Section 4.3 is a highly specific physics term that, while necessary for the case study, should be contextualized more clearly for non-specialists if the paper aims for broad impact. These terms should be defined at first use or replaced with plainer alternatives to ensure the paper is accessible to the full spectrum of scientific researchers.

**Action items.**

- [**writing**] The manuscript relies heavily on domain-specific acronyms and jargon that hinder accessibility for a broad scientific audience. In Section 3.2, the term “ReAct-style loop” is introduced without definition; while standard in AI agent literature, it should be briefly explained (e.g., “a reasoning-and-action loop”) for general readers. Similarly, Section 5.4 utilizes “XEB” (Cross-Entropy Benchmark) without expansion, which is opaque to non-quantum computing specialists. The term “dry-lab” in Sectio

## Logical Consistency

*Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a logical inconsistency between its evaluation metric design and its primary conclusion regarding “discovery.” In Section 3.3, the authors define a score boundary of 50, where scores >50 indicate “new discovery.” However, Section 3.1 and the Appendix explicitly state that the evaluation rubrics are constructed from “hidden target papers” ($p^\star$). If the ground truth for evaluation is derived entirely from existing literature, the rubric inherently caps the maximum possible score at the level of the target paper’s known contributions. Logically, a system cannot generate a score exceeding the rubric’s definition of the target to prove “new discovery” unless the rubric includes criteria for novelty that are independent of the target paper. The paper fails to explain how the scoring mechanism distinguishes between a perfect reproduction (50) and a genuine new finding (>50) when the evaluation artifacts ($\mathcal{A}$) are anchored to the target.

Furthermore, the claim that current agents are “far from reliable re-discovery” (Introduction) rests on the assumption that a score of 50 equates to a successful re-discovery. The rubrics are weighted sums of specific artifacts (e.g., 0.20 for ingestion, 0.30 for exclusion curves). The paper does not provide a logical derivation or empirical validation that the sum of these specific weighted components is sufficient to constitute a “match” to the target paper’s overall scientific validity. Without this link, the conclusion that low scores (e.g., 21.5) represent a failure to “re-discover” is an overreach; they may simply represent a failure to meet specific, potentially arbitrary, rubric weights.

Finally, the error analysis in Section 4.4 categorizes failures as “Experimental Protocol Mismatch.” However, the case study for Physics_002 (Section 4.5) describes an agent that successfully recovered the "direct XEB trend" (a correct protocol for that specific observation) but failed to perform "verification steps" (log-XEB, mirror-circuit inference). This behavior is logically better described as "Scope Incompleteness" or "Partial Execution" rather than a "Mismatch," as the agent did not necessarily choose the wrong protocol, but rather an incomplete one. This misclassification weakens the causal claim that agents are fundamentally misaligned with scientific protocols.

**Action items.**

- **[writing]** The paper presents a logical inconsistency between its evaluation metric design and its primary conclusion regarding "discovery." In Section 3.3, the authors define a score boundary of 50, where scores >50 indicate "new discovery." However, Section 3.1 and the Appendix explicitly state that the evaluation rubrics are constructed from "hidden target papers" ($p^{\star}$). If the ground truth for evaluation is derived entirely from existing literature, the rubric inherently caps the maximum possible

## Overreach

*Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_overreach.md`

The paper makes a significant overreach regarding the interpretation of its scoring metric. In the Introduction and Section 3.3, the authors explicitly state that "scores above 50 indicate new discovery." This claim is not supported by the data or the methodology described. The evaluation rubrics are constructed by domain experts based on the *hidden target paper* (the ground truth). Consequently, the scoring mechanism is designed to measure how closely a system's output matches the known target. A score of 50 represents a perfect match to the target. There is no evidence in the paper that the rubrics can objectively validate a *novel* scientific finding that differs from the target paper. If a system were to produce a genuinely new discovery, the rubric—anchored to the old paper—would likely penalize it for deviating from the target, or the "discovery" would be unverified by the current evaluation framework. The claim that the benchmark can currently measure "discovery" is an extrapolation beyond the capabilities of the described rubric-based evaluation.

Furthermore, the paper over-generalizes its findings to the broader field of "autonomous scientific research." While the benchmark is ambitious, the "Limitations" section admits it is restricted to "dry-lab" tasks and does not cover wet-lab operations or instrument control. The conclusion that "current systems are far from reliable scientific re-discovery" is valid for the specific scope of the benchmark, but the framing suggests a broader failure of AI in science that the data does not fully support. The authors should clarify that the benchmark evaluates *computational re-discovery* of existing results from raw data, rather than the full spectrum of scientific research which includes experimental design in physical labs and truly novel hypothesis generation.

Finally, the claim that the benchmark spans “10 scientific domains” (Introduction) is technically true based on the task list, but the depth of coverage in each domain is minimal (often just 4 tasks per domain). The paper implies a comprehensive evaluation across these fields, but the small sample size per domain limits the statistical power of any domain-specific conclusions. The authors should avoid implying that the results are representative of AI capabilities across the entire breadth of these 10 fields.

**Action items.**

- [**writing**] The paper makes a significant overreach regarding the interpretation of its scoring metric. In the Introduction and Section 3.3, the authors explicitly state that “scores above 50 indicate new discovery.” This claim is not supported by the data or the methodology described. The evaluation rubrics are constructed by domain experts based on the *hidden target paper* (the ground truth). Consequently, the scoring mechanism is designed to measure how closely a system’s output matches the known target

## Safety Ethics

*Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a significant benchmark for autonomous scientific research, but several safety and ethical considerations require clarification before publication.

First, the evaluation methodology relies heavily on an LLM (GPT-5.1) to act as the judge for 280 runs across 40 tasks (Section 4.1, “Experimental Setup”). While the paper mentions “expert-curated rubrics,” the actual scoring is automated by another AI. This introduces a risk of “judge bias” or “hallucinated compliance,” where the judging model might favor outputs that sound plausible but are scientifically incorrect, or conversely, penalize valid novel approaches that deviate from the rubric’s phrasing. Given the paper’s conclusion that current systems are “far from reliable,” the authors must provide evidence that the judging model is not systematically misinterpreting the rubrics or introducing its own biases. A section detailing the validation of the judging model (e.g., human spot-checks, inter-rater reliability with human experts) is necessary to ensure the safety and validity of the benchmark’s conclusions.

Second, the dataset includes tasks related to human biology and health, specifically `Life_001` (neoantigen vaccine design using patient-specific sequencing, HLA typing, VAF) and `Life_003` (nanopore signal data). The manuscript does not explicitly state the provenance of this data or the ethical safeguards employed. The authors must confirm that all patient data used in the benchmark is either fully synthetic, aggregated to the point of non-reidentifiability, or derived from public datasets with appropriate Data Use Agreements (DUAs) and IRB approvals. If any real patient data was used, the lack of an explicit statement regarding consent and privacy compliance is a critical ethical gap.

Third, the “ResearchHarness” environment provides agents with tools for web search, file manipulation, and code execution (Section 3.2). While the tasks are "dry-lab," the unrestricted nature of these tools poses a potential dual-use risk if the benchmark were to be extended or if agents were to "jailbreak" the constraints to access restricted databases or generate harmful protocols (e.g., synthesizing pathogens). The authors should briefly describe the safety guardrails, sandboxing, or monitoring mechanisms in place to prevent agents from executing actions outside the defined scientific scope or accessing sensitive information during the evaluation.

Finally, the paper lists "qwen.qwen3.5-122b" as one of the paper authors. This is an unusual inclusion for a human-authored manuscript and raises questions about the transparency of AI contribution. While not a direct safety violation, the authors should clarify the role of this entity to ensure compliance with authorship guidelines regarding AI tools versus human contributors.

**Action items.**

- [**science**] The paper relies on GPT-5.1 to score 280 agent runs against rubrics (Section 4.1). Given the high stakes of evaluating scientific discovery, the authors must explicitly address the risk of LLM-as-a-judge bias, hallucination in rubric adherence, and potential conflicts of interest if the judging model shares lineage with the evaluated agents. A human-in-the-loop validation sample or inter-rater reliability metric is required.
- [**science**] The benchmark includes tasks involving human health data (e.g., Life_001 neoantigen vaccines, Life_003 nanopore sequencing). The manuscript must confirm that all patient-specific sequencing and HLA typing data used in these tasks are fully anonymized, aggregated, or synthetic, and that no IRB approval or data use agreement violations occurred during the construction of the benchmark dataset.
- [**writing**] The evaluation harness grants agents unrestricted web access (WebSearch, ScholarSearch) and code execution (Bash, Terminal). The authors must describe the safety guardrails implemented to prevent agents from accessing non-public repositories, executing harmful commands, or generating dual-use biological/chemical protocols during the benchmark runs.

## Scientific Evidence

*Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a novel benchmark for end-to-end autonomous scientific research, but the strength of the scientific evidence supporting its central claims is currently insufficient due to methodological gaps in evaluation and statistical rigor.

First, the primary metric—the rubric score—is entirely dependent on an automated judge (GPT-5.1) as stated in Section 4.1 ("Scoring is performed by GPT-5.1 against rubrics"). The manuscript provides no evidence that this automated judge correlates with human expert judgment or ground-truth scientific validity. Without a validation study (e.g., inter-rater reliability with human domain experts, or a correlation analysis against known correct solutions), the quantitative results (e.g., Claude Code scoring 21.5 vs. Claude-Opus-4.7 at 20.7) are scientifically unverified. The difference of 0.8 points is well within the likely noise floor of an uncalibrated LLM judge, rendering the ranking of agents potentially spurious.

Second, the sample size of 40 tasks (approximately 4 per domain) is statistically underpowered to support broad claims about performance across "10 scientific domains" (Introduction). The paper reports mean scores but fails to provide standard deviations, confidence intervals, or variance breakdowns per task or per domain. This omission prevents the reader from assessing whether the low scores are consistent failures or driven by a few outlier tasks. A power analysis or at least a discussion of the statistical significance of the observed differences is necessary to validate the conclusion that "current systems are far from reliable."

Finally, the error analysis in Section 4.4 aggregates 280 runs to identify failure modes. However, without reporting the distribution of scores or the variance within each error category, the claim that failures "concentrate" on specific types (e.g., "Experiment Design Mismatch") lacks statistical backing. It is unclear if these patterns are robust or artifacts of the small sample size. The paper must include statistical tests (e.g., chi-square for error distribution) and report confidence intervals for the mean scores to ensure the conclusions are robust to plausible alternative explanations regarding task difficulty or judge bias.

**Action items.**

- [**science**] The central claim relies on GPT-5.1 as an automated judge for rubric scoring (Section 4.1). The paper lacks a validation study (e.g., inter-rater reliability with human experts or correlation with ground-truth outcomes) to prove the judge's scores are robust and not hallucinated. Without this, the quantitative results (e.g., 21.5 vs 20.7) are scientifically unverified.

- [**science**] The sample size of 40 tasks across 10 domains (4 tasks/domain) is statistically underpowered for drawing broad conclusions about '10 scientific domains' (Introduction). The variance within domains is not reported, making it impossible to determine if the low scores are due to model failure or specific task difficulty. A power analysis or confidence intervals for the mean scores are required.

- [**science**] The error analysis aggregates 280 runs (7 agents x 40 tasks) but does not report the variance or standard deviation of scores per task. The claim that 'failures concentrate' on specific error types (Section 4.4) requires statistical testing (e.g., chi-square) to distinguish signal from noise, especially given the small number of tasks per domain.

## Statistical Analysis

*Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_statistical_analysis.md`

The manuscript presents a comprehensive benchmark for autonomous scientific research, but the statistical rigor of the reported results requires significant strengthening before the claims can be fully accepted.

First, the primary results section (Section 4.2) reports mean scores for various agents and LLMs (e.g., Claude Code at 21.5, LLM Frontier Mean at 26.5) without any measure of statistical uncertainty. With only 40 tasks, the standard error of the mean is non-negligible, and the variance across domains (Astronomy vs. Chemistry) appears high. The authors must report 95% confidence intervals for all aggregate scores. Furthermore, claims regarding the relative performance of systems (e.g., "Claude Code leads agents") require formal statistical testing (e.g., paired t-tests or non-parametric equivalents like the Wilcoxon signed-rank test) to determine if observed differences are statistically significant or merely artifacts of the specific task sample.

Second, the evaluation methodology relies on an LLM (GPT-5.1) to score the outputs against rubrics. The paper does not provide a statistical validation of this "judge." There is no report of inter-rater reliability (e.g., Cohen's kappa or intraclass correlation) comparing the LLM judge's scores against human expert scores, nor is there an analysis of the judge's stability (e.g., via bootstrapping or multiple runs). Without establishing the reliability and variance of the scoring mechanism itself, the reported scores lack a statistical foundation.

Finally, the error analysis (Section 4.5) aggregates 280 runs into error categories but presents these as raw counts or percentages without confidence intervals. To support the claim that failures "concentrate" on specific types (e.g., Experiment Design Mismatch), the authors should provide confidence intervals for these proportions and test whether the distribution of errors differs significantly across different agent architectures.

Addressing these points is essential to ensure the benchmark's results are robust and reproducible.

**Action items.**

- **[science]** The paper reports aggregate scores (e.g., 21.5, 26.5) and error rates without providing measures of statistical uncertainty (standard error, 95% confidence intervals) or significance tests. Given the small sample size (N=40 tasks) and high variance across domains, the authors must report confidence intervals for all mean scores and perform appropriate statistical tests (e.g., paired t-tests or Wilcoxon signed-rank) to validate claims of superiority between systems.

- **[science]** The evaluation relies on an LLM-as-a-judge (GPT-5.1) to score rubrics. The manuscript lacks a statistical validation of this scoring mechanism, such as inter-rater reliability (Cohen's kappa) against human experts or a stability analysis (e.g., bootstrapping) to quantify the variance introduced by the judge model. Without this, the reliability of the reported scores is statistically unverified.

- **[science]** The error analysis aggregates 280 runs into six categories but does not report confidence intervals for these proportions or test for statistical significance in the distribution of errors across different agents. Claims about "concentration" of failures need statistical backing to distinguish signal from noise.

## Writing Quality

*Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high level of technical sophistication and presents a complex benchmark with clear structure. However, there are specific instances where the prose becomes dense or slightly ambiguous, which impedes the immediate clarity of the argument.

In Section 5.2, the analysis of the relationship between resource investment and score contains a confusing sentence: "this relationship is largely elevated by Claude Code." The verb "elevated" is imprecise here; it is unclear if the author means the correlation is "driven," "skewed," or "artificially inflated" by this single data point. Given the context of Pareto frontiers and outliers, a more precise term is required to ensure the reader understands that Claude Code is an anomaly affecting the statistical trend.

Similarly, in Section 5.3, the distinction between "scientific goal misalignment" and "insufficient iterative trial-and-error" is slightly muddled. The phrasing suggests that "insufficient iterative trial-and-error" is a category of failure comparable to misalignment, whereas the intended meaning is likely that the *lack* of sufficient iteration is not the primary cause of failure. A minor syntactic adjustment would clarify that the failures are due to misalignment rather than a lack of effort.

Finally, the Appendix (specifically the `Dataset Summaries` in e001) suffers from formatting issues that affect readability. The entry for `Life_001` lists numerous CSV filenames in a single, dense block of text. This makes it difficult for the reader to parse the specific data components. Converting these lists into structured bullet points or a small table would significantly improve the flow and usability of the appendix without altering the scientific content.

Overall, the writing is professional, but these specific areas require polishing to ensure the complex findings are communicated with maximum clarity.

**Action items.**

- [**writing**] In Section 5.2 (Runtime and Cost Analysis), the phrase 'this relationship is largely elevated by Claude Code' is semantically unclear. 'Elevated' typically implies raising a value, but the context suggests the relationship is 'driven' or 'skewed' by this specific data point. Rephrase to clarify that Claude Code is an outlier driving the correlation.
- [**writing**] In Section 5.3 (Error Analysis), the sentence 'system failures more often reflect scientific goal misalignment... than insufficient iterative trial-and-error' is slightly ambiguous. It is unclear if 'insufficient iterative trial-and-error' is a type of failure or a cause. Consider rephrasing to: 'failures stem more from scientific goal misalignment... than from a lack of iterative trial-and-error.'
- [**writing**] In the Appendix (e001), the dataset summary for `Life_001` contains a long, unbroken list of filenames that disrupts readability. Consider using a bulleted list or a table to present the input/output files clearly, rather than a dense paragraph.